Short Diophantine Consistency Equations: A Squared-Modulus and Root-Gap Synthesis

Arithmetic research packet

23 October 2026

Abstract

Combining three previously established coordinate reductions gives nine new local fifteen-positive-variable degree/operation tradeoffs and lowers the degree of a fourteen-variable, 79-operation equation from 56 to 52. The inherited absolute operation record remains 66; the inherited degree-44 point remains unchanged. We give exact literal polynomials, positive existence reductions, coherent materialized coefficients and independent finite arithmetic audits, while explicitly retaining the inherited compiler proof boundary.

1 A synthesized short-polynomial frontier

The squared-modulus quotient, ordinary strong-rank theorem, and two Pell-root gaps are all inherited. Their combination improves several local degree/operation tradeoffs, without reducing the absolute operation record below the incoming 66-operation, degree-157 equation. All equations below have strictly positive integer unknowns, and their lack of solutions is provably equivalent in ZFC to Con(ZFC) at the ordinary mathematical proof level of the inherited compiler. This is not a new end-to-end Lean/Rocq certificate or a consistency assertion.

Variables	Operations	Exact degree	Construction
15	67	125	Reduced scaled product; squared modulus
15	68	101	Ordinary product; squared modulus
15	69	91	Ordinary product; first gap
15	70	84	Ordinary product; both gaps
15	71	80	Three squared residuals
15	72	74	Three squared residuals; first gap
15	73	60	Four squared residuals
15	74	50	Four squared residuals; auxiliary gap
15	75	46	Four squared residuals; both gaps
14	79	52	Absorber; squared modulus; auxiliary gap

Comparison discipline. The incoming 15-variable 66/157 and 76/44 points remain incomparable endpoints. Historical pairs 67/155, 68/123, 69/99, 70/93, 71/83, 74/56 and 78/44, together with incoming 68/113 and wave-7 72/80, are retained as comparators. The 14-variable point improves the incoming 79/56 construction at the same operation count. The included 76/44 reconstruction is a reference control, not new work. These are bounded repository comparisons, not global minimality or priority claims.

1.1 Literal fixed polynomials

Use the entire Wang943 121×8801 tuple, with $B = 2^{1064921}$, $\beta = B - 1$, $K_0 = K$, $M_C = M_C$, and $M_F = M_{F,\text{source}} = M_{F,\text{native}} + B - 1$. All four materialized coefficients and their compressed/raw hashes

accompany the report. In particular, M_F is already shifted. For fifteen coordinates use

$$J, F, \alpha, z, f, h, i, T, s, w, \tau, \eta, \zeta, y, \gamma > 0.$$

Replace τ by r when the first gap is enabled and y by u when the auxiliary gap is enabled. Define polynomial abbreviations

$$\begin{aligned}
&q = \beta J + 1, && C = q - F - \alpha, \quad X = wq, && Y = sq^3,\\
&E = XY, && k = \eta + \zeta, \quad c = kY + \eta, && a = E + Y,\\
&H = 4a + 3, && \Delta = a^2 + H, \quad D = X + ac + \gamma H, && L = EkY,\\
&R = ((q-1)(q-F) + \alpha)(q^2 - 1) + (M_C + qM_F)J,\\
&V = f^2(cT - R) - c, && A_0 = \Delta(f^2 - 1), \quad I = ic^2.
\end{aligned}$$

They introduce no additional unknowns. The norm factors are

$$\begin{aligned}
&N_0 = \tau^2 - L(L + k), && N_1 = D^2 - \Delta c^2,\\
&N_a = A_0(V^2 - y^2) + y^2, && N_k = k - hE - R,\\
&N_t = (K_0 + w)C + (q - F) - z(q - 1), && S = I^2 - A_0, \quad N_s = S + 1.
\end{aligned}$$

The gap alternatives are

$$N_{0,g} = r^2 + L(2r - k), \qquad N_{a,g} = (V + u)^2 - A_0 u(2V + u).$$

Let $\mathcal{Q}(B_1, \ldots, B_m) = S^2 + \sum_j (\prod_{N \in B_j} N - 1)^2$. The displayed data specify the ordinary polynomials completely:

Operations	Equation's left side	Enabled gaps
68	$N_0N_1N_aN_kN_tN_s - 1$	None
69	$N_0N_1N_aN_kN_tN_s - 1$	First
70	$N_0N_1N_aN_kN_tN_s - 1$	Both
71	$\mathcal{Q}(\{N_0, N_1\}, \{N_a, N_k, N_t\})$	None
72	$\mathcal{Q}(\{N_0, N_1, N_k\}, \{N_a, N_t\})$	First
73	$\mathcal{Q}(\{N_0\}, \{N_a\}, \{N_1, N_k, N_t\})$	None
74	$\mathcal{Q}(\{N_a\}, \{N_1, N_k\}, \{N_0, N_t\})$	Auxiliary
75	$\mathcal{Q}(\{N_a\}, \{N_0, N_k\}, \{N_1, N_t\})$	Both

For the 67-operation equation only, put $I_{\mathrm{sc}} = i\Delta c^2$ and $N_{\mathrm{sc}} = \Delta f^2 - I_{\mathrm{sc}}^2$. With no gaps, its left side is

$$N_0N_1N_aN_kN_tN_{\mathrm{sc}} - \Delta.$$

The evaluator shares Δf^2 between N_{sc} and $A_0 = \Delta f^2 - \Delta$. Every binary operation, including fixed-scalar multiplication, is charged in the literal JSON gate lists.

1.2 Why the combinations preserve existence

The exact all-value pullback is

$$c((fT)f - 1) - Rf^2 = f^2(cT - R) - c.$$

Thus a new positive quotient gives the old positive quotient $T_{\mathrm{old}} = fT$. Completeness is not the assertion that every old quotient is divisible by f. Instead, at a canonical compiler witness, write $A = a + 2$ and $c = \psi_A(R)$, where $R \equiv 3 \pmod 4$. Ordinary strong rank supplies

$$I^2 = \Delta(f^2 - 1), \qquad c \mid I, \qquad f^2 \equiv 1 \pmod c.$$

For $R = 2j + 1$ the integer polynomial Q_j defined by $\chi_t(R) = tQ_j(t^2)$ satisfies

$$Q_j(0) = (-1)^j R, \qquad Q_j(1 - A^2) = (-1)^j \psi_A(R).$$

The recurrence $Q_{j+1} = (4Z - 2)Q_j - Q_{j-1}$, with $Q_0 = 1$ and $Q_1 = 4Z - 3$, proves both identities. Since j is odd, set

$$V_* = \frac{\chi_I(R)}{I}, \qquad y_* = \psi_I(R).$$

The quotient is integral, $V_* \equiv -c \pmod{f^2}$ and $V_* \equiv -R \pmod{c}$. Since $\gcd(c, f) = 1$, the number

$$T_* = \frac{V_* + c + Rf^2}{cf^2}$$

is a positive integer and satisfies the new quotient formula. The auxiliary norm remains one. This argument works with the ordinary strong coordinate; it never divides an arbitrary supplied i by Δ.

The root gaps use exact polynomial identities:

$$(L + r)^2 - L(L + k) = r^2 + L(2r - k),$$
$$A_0\big(V^2 - (V + u)^2\big) + (V + u)^2 = (V + u)^2 - A_0 u(2V + u).$$

For soundness restore $\tau = L + r > 0$ and $y = |V + u|$. Once $A_0 = I^2 > 1$, an auxiliary unit excludes $V + u = 0$. For completeness $r = \tau - L > 0$ follows from $\tau^2 - L^2 = Lk + 1 > 0$. The canonical $V_* > 1$ and

$$(I^2 - 1)(y_*^2 - V_*^2) = V_*^2 - 1 > 0$$

give $u = y_* - V_* > 0$.

For the ordinary product, all factors are units. The full mod-four check shows $N_s = 1 + I^2 - \Delta(f^2 - 1) \not\equiv -1 \pmod 4$, since $\Delta \equiv 0$ or 3 and squares are 0 or 1. Hence $S = 0$; in the squared sums this is immediate. The consecutive-product negative Pell obstruction excludes $N_0 = -1$; modulo four excludes $N_1 = -1$ and, now that $A_0 = I^2$, $N_a = -1$. Thus

$$N_0 = N_1 = N_a = 1, \qquad N_k N_t = 1, \qquad S = 0.$$

Every grouping accounts for each of the five factors once, so these values pull back to the inherited relaxed 72-operation polynomial. Its proof retains the possible $C = 0$ boundary, establishes the large main index before invoking ordinary rank, recovers $p = R$, and only then forces $N_k = N_t = 1$ and $C > 0$. The authenticated native compiler and blank-K215 calibration supply the finite ZFC refutation. Conversely, a canonical compiler witness makes all individual factors one after the replacements.

For the scaled equation cancel positive Δ and apply the normalized strong-unit mod-four obstruction. Then $N_{\mathrm{sc}} = \Delta$ and $A_0 = I_{\mathrm{sc}}^2$, reducing to the inherited scaled proof. The canonical completeness argument above applies with I_{sc} in place of I. Full local arguments and inherited interfaces are included in `PROOF.md` and the authenticated source-proof directory.

1.3 The fourteen-variable degree reduction

Remove h, retain τ, and replace y by u. Put

$$U = 4q^4, \quad r_0 = k - R - 1, \quad b_0 = X + ac, \quad D_* = Ub_0 + H(U\gamma - r_0), \quad G_* = 1 + \Delta c^2.$$

The equation is

$$(D_*^2 - U^2 G_*)^2 + S^2 + (N_{a,g} - 1)^2 + (N_0 N_t - 1)^2 = 0.$$

It is an exact existence adapter of the incoming 14-variable 79/56 absorber: $T_{\mathrm{old}} = fT$ and $y = |V + u|$ give soundness directly. The absorber proof's early index bound and ordinary rank theorem are inherited,

not replaced. Canonical squared-modulus reconstruction followed by $u = y_* - V_*$ supplies completeness. One saved quotient gate pays for one added auxiliary-gap gate, leaving 79 operations.

The exact main residual identity is

$$\begin{aligned} D_*^2 - U^2 G_* = U^2((b_0 + H\gamma)^2 - G_*) \\ - 2UHr_0(b_0 + H\gamma) + H^2 r_0^2. \end{aligned}$$

Its three terms have degrees $26, \leq 25, \leq 20$; the other residuals have degrees $22, 23, 24$. Therefore the exact final degree is 52, not 44. With $Q = \beta J$ and $k_0 = \eta + \zeta$, its nonzero leading form is $16384\gamma^2 w^4 s^6 k_0^2 Q^{38}$.

1.4 Arithmetic verification and proof boundary

The independent standard-library auditor imports no generator or upstream Python code. It expands each literal main-norm ancestor cone exactly, then propagates symbolic leading forms, preserving cancellations rather than using only naive degree bounds. The factor degrees are

$$(\deg N_0, \deg N_1, \deg N_a, \deg N_k, \deg N_t, \deg S) = (22 \text{ or } 12, 18, 30 \text{ or } 23, 7, 2, 22).$$

The scaled factor has degree 46. All output leading forms have an exhibited monomial with nonzero coefficient depending only on $\beta > 0$, so the proof applies to the actual bound tuple. All gates, unknowns and ports are live. Raw/compressed coefficient hashes, ranges, mask populations and source-proof hashes are checked, along with five universal local polynomial identities. Each of the eleven circuits passes 100 signed/degenerate literal-formula diagnostics; five exact canonical Pell examples check reconstruction and positive gaps. Checks pass normally and under `python3 -O`.

These are finite artifact checks, separate from the unbounded Pell and compiler proofs. No finite experiment tests Con(ZFC). The contribution is the explicit combined frontier, reproducible circuits and reduction proof, not a claim of a foundational breakthrough.